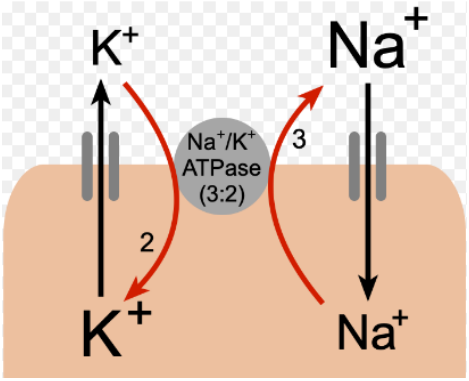
Ion balance: content

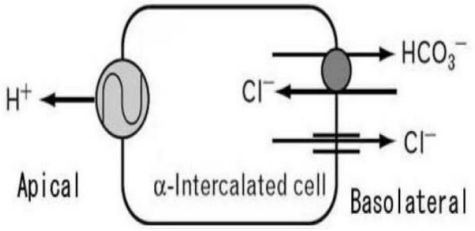
- K^+: ECG
- Na^+: osmolarity
- Ca^{2+}: tetany and renal osteodystrophy
  - Parathyroid hormone, vitamin D and calcitonin
- Phosphates: Renal osteodystrophy
- Mg^{2+}:

Approximate major electrolyte content in body fluid

Extracellular fluid: plasma		Intracellular fluid	
Electrolytes	mmol/L	Electrolytes	mmol/L
Cations		Cations	
Na^+	135-145	K^+	150
K^+	3.5-5.5	Mg^+	40
Ca^+	5	Na^+	10
Mg^+	2		
Anions		Anions	
Cl^-	103	Phosphates and sulfates	150
HCO_3^-	26	HCO_3^-	10
HPO_4^{2-}	2	Proteinate	40
HSO_4^-	1		
Organic acids	5		
Proteinate	17		



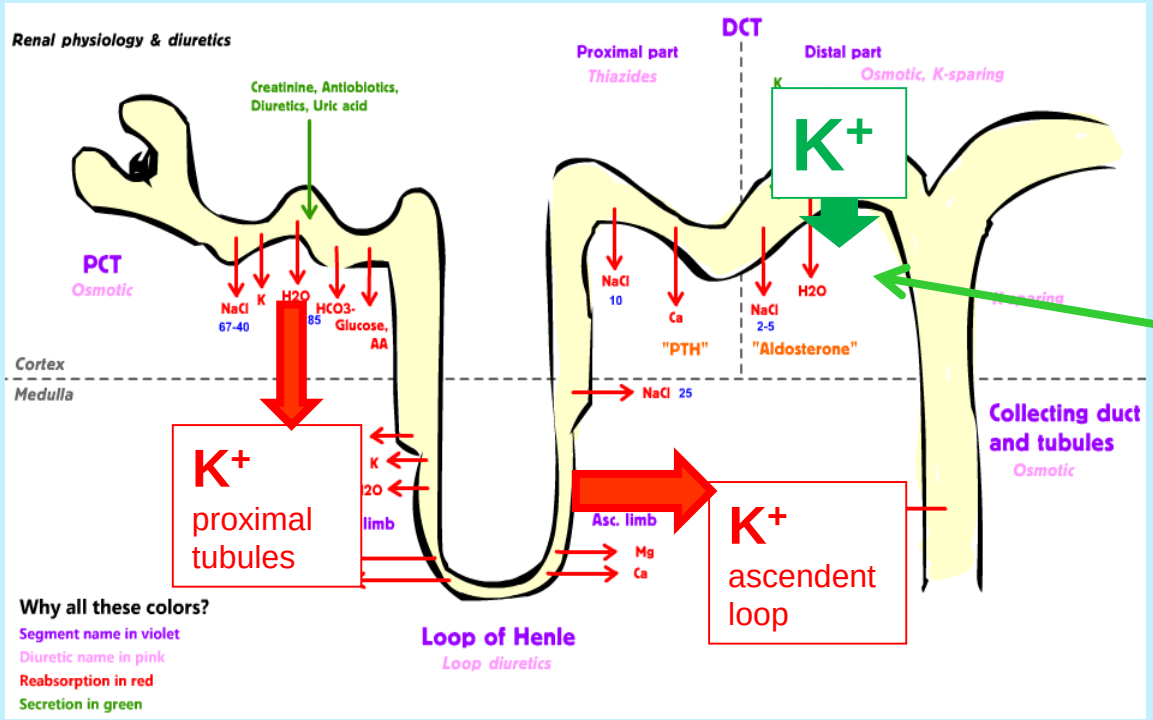
3Na+/2K+ ATPase pump



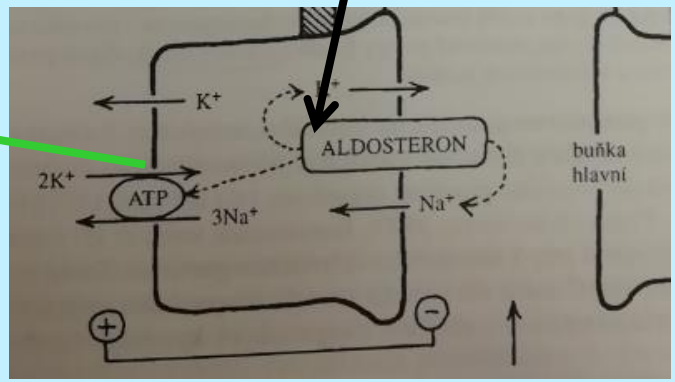
K^+

K^+: a role of aldosterone

- Renal excretion of K^+ is regulated by aldosterone that is stimulated by RAAS, ↑ blood K^+ and ↑ Na^+ in urine.
  - Aldosterone activates Na^+/K^+ ATPase pump on basolateral membrane. It ↑ intracellular K^+ that is excreted to urine. It also ↓ intracellular Na^+ that is replaced by Na^+ from urine.



↑ urinary Na^+



The distal tubules

↑K^+ Hyperkalemia (above 5.5 mmol/L): causes

- Intra-extracellular shift:
  - Acidemia but final excretion leads to hypokalemia
  - Low insulin
- Rhabdomyolysis
- Decreased renal loss
  - Hypocortisolism: Addison disease (low aldosterone)
  - Hyporeninemic hypoaldosteronism in diabetic nephropathy or interstitial nephritis
  - Acetylsalicylic acid that inhibits prostaglandin and renin
  - Kidney failure in last stage. Chronic renal insufficiency (K^+ shift from intracellular space to extracellular space and the kidney cannot excrete K^+)

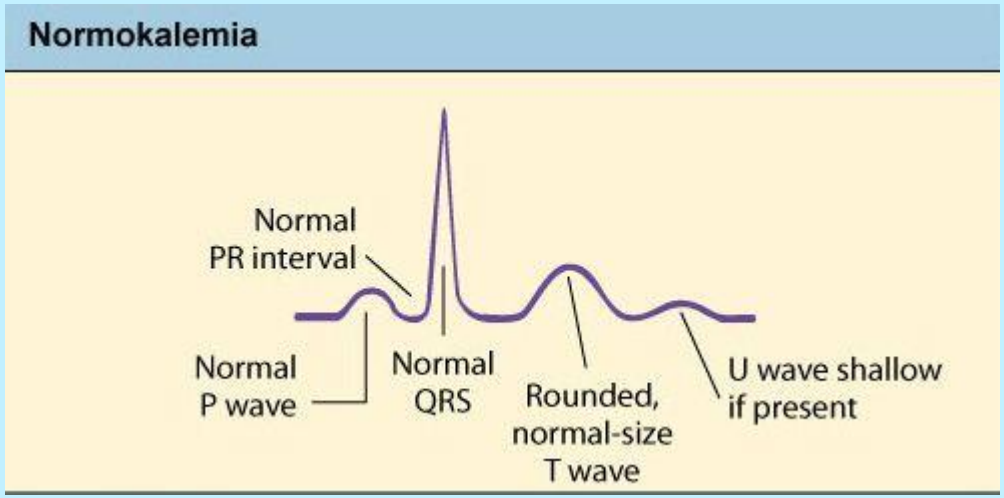
↑K^+ Hyperkalemia (above 5.5 mmol/L): manifestation

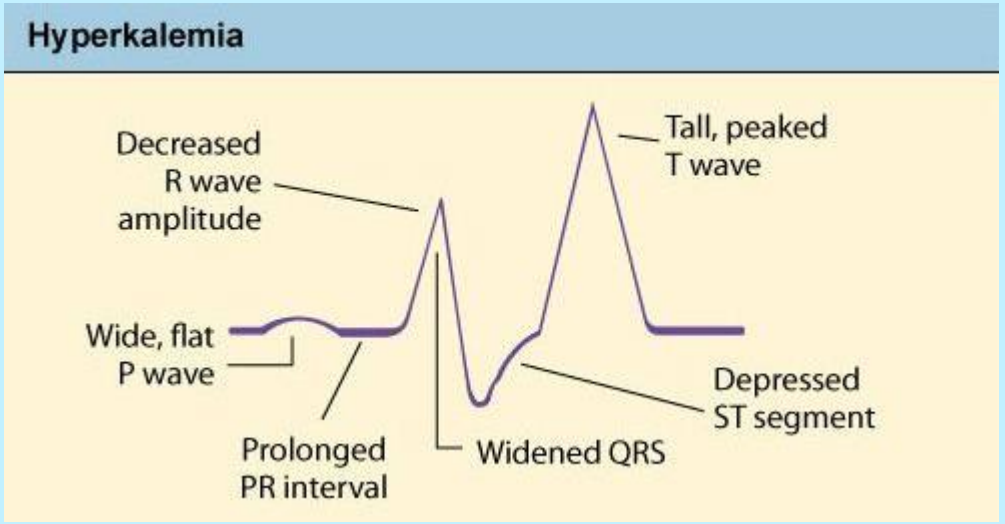
- Neuromuscular : muscle weakness, muscle twitching, paresthesia (prickling sensation)

- ECG: **Bradyarrhythmia:** sinus bradycardia, high-grade AV block with slow junctional and ventricular escape rhythms

↑K^+ ECG hyperkalemia-fast repolarization

- Potassium: above 5.5 mmol/L





↑K^+ ECG hyperkalemia: stages

- Mild hyperkalemia may not be associated with any acute symptoms and vital signs may be normal.
- Moderate hyperkalemia may present clinically with acute extremity paresthesia and bradyarrhythmias (↑K^+ decreases Na^+ influx), e.g.sinus bradycardia.

Serum potassium	Typical ECG appearance	Possible ECG abnormalities
Mild (5.5–6.5 mEq/L)		Peaked T waves Prolonged PR segment
Moderate (6.5–8.0 mEq/L)		Loss of P wave Prolonged QRS complex ST-segment elevation Ectopic beats and escape rhythms
Severe (>8.0 mEq/L)		Progressive widening of QRS complex Sine wave Ventricular fibrillation Asystole Axis deviations Bundle branch blocks Fascicular blocks

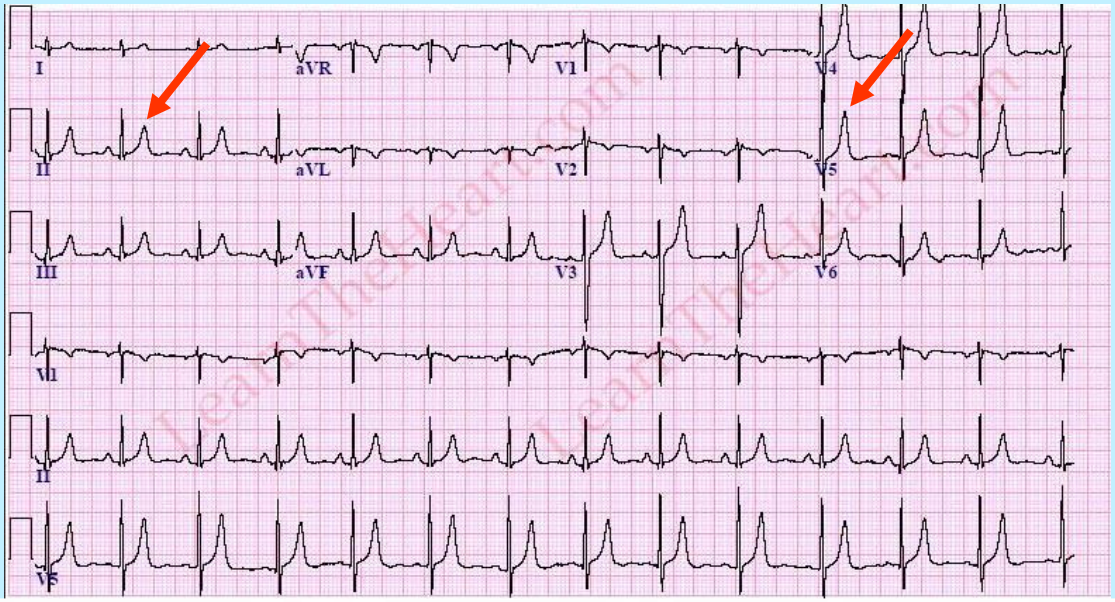
EpoMedicine

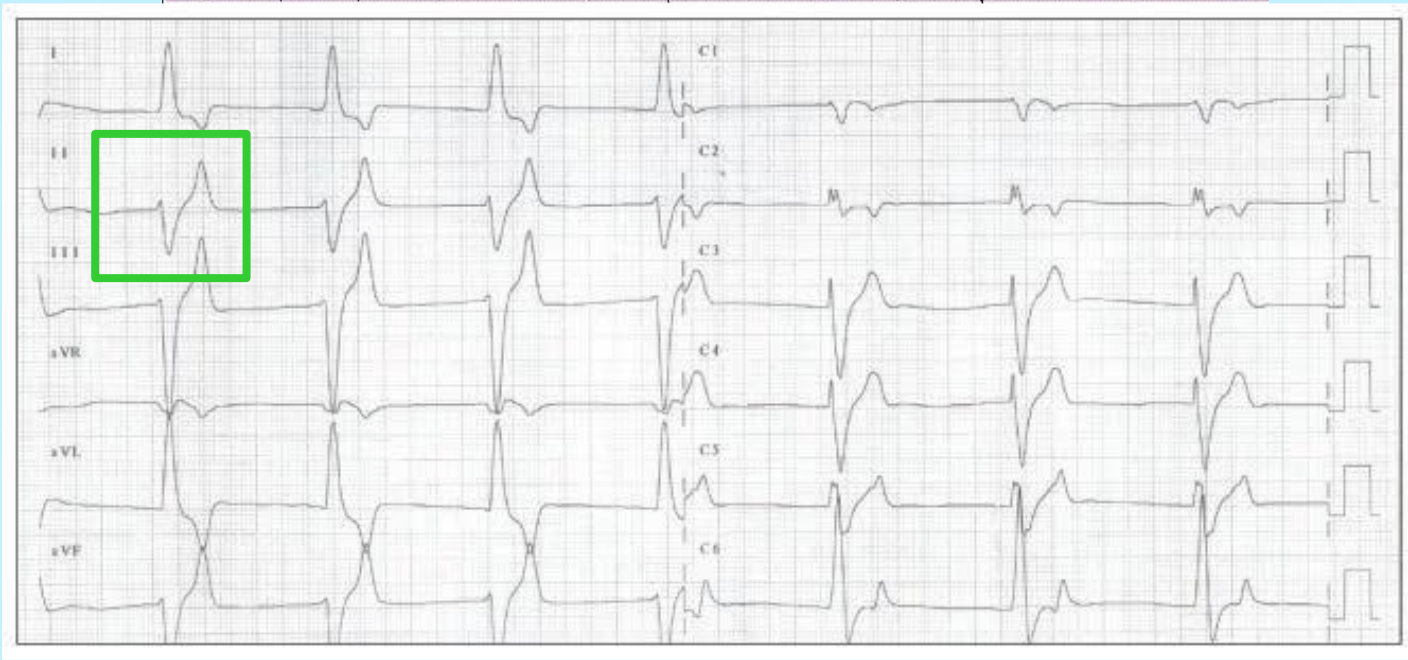
- Severe hyperkalemia:
  - Sine wave appearance
  - Ventricular fibrillation
  - Asystole

Sine wave = QRS and T wave fuse each other

↑K^+

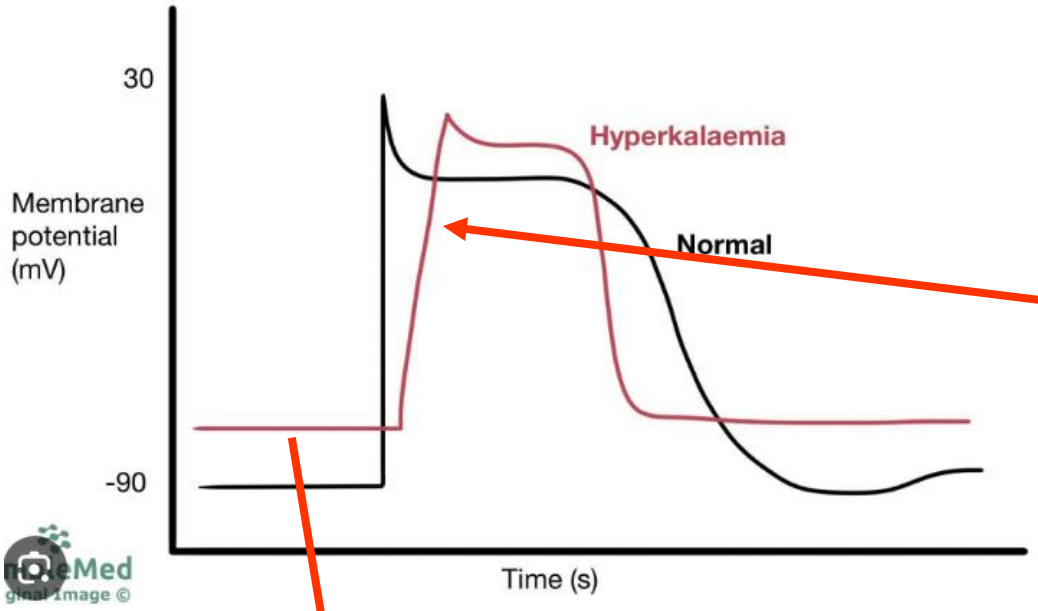
ECG hyperkalemia





↑K^+

ECG hyperkalemia - bradyarrhythmia



small Na+ enter

- The decreased resting membrane potential which decreases Na+ influx can lead to bradyarrhythmia, e.g. sinus bradycardia

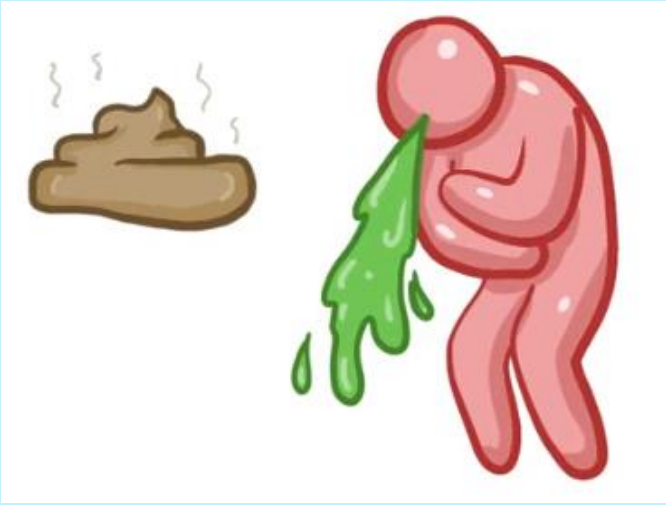
↓K^+ Hypokalemia (below 3.5 mmol/L): causes

- Intra-extracellular shift:
  - Alkalemia (acidemia)
  - High insulin and adrenalin (pheochromocytoma)
  - Massive catabolism of intracellular proteins and glycogen: decreased storage
- Increased renal loss
  - Hyperaldosteronism
  - Hyperreninism in renal arterial stenosis
  - ↑ Na^+ availability in the renal tubules
    - Type I and Type II renal tubular acidosis
    - Bartter syndrome and Loop diuretics (in ascendent loop Na^+-K^+-$2Cl^-$ symporter) leading to alkalosis
    - Thiazide diuretics(in distal tubules Na^+Cl^- symporter) : alkalosis
- Extrarenal K^+ losses from the body are usually small, but can be marked in chronic diarrhea, severe burns or prolonged sweating
  - Diarrhea: Loss in GIT leads to hyperchloremic metabolic acidosis with ↓K+
  - Vomiting leading to chypochloremic metabolic alkalosis
  - An abuse of laxatives and the use of diuretics- both loop and thiazides.

↓K^+ Hypokalemia (below 3.5 mmol/L): manifestation

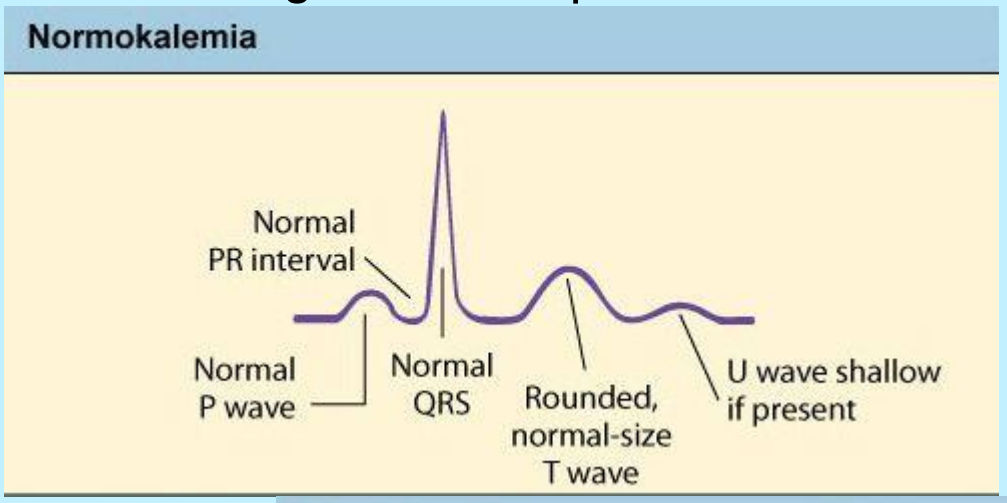
- Neuromuscular : smooth muscle weakness leading to obstipation, paralytic ileus or even paralysis of respiratory muscle.

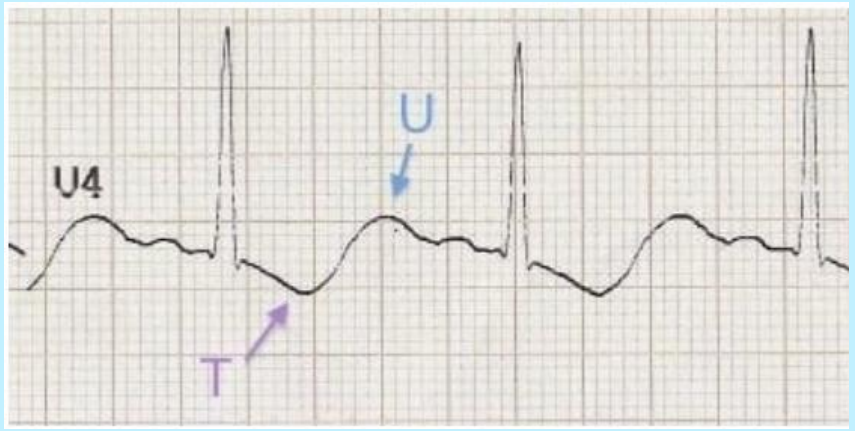
- ECG: tachyarrhythmias with palpitation

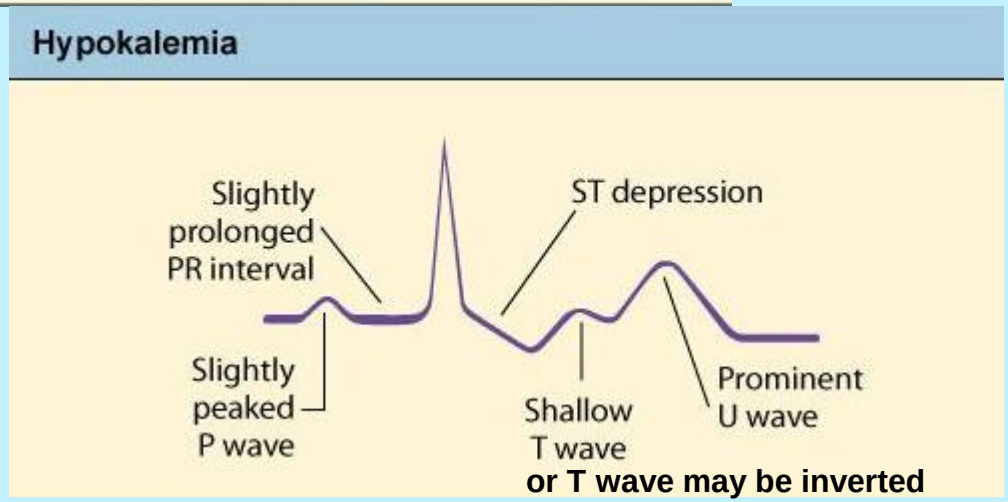


↓K^+ ECG hypokalemia-delayed repolarization

- Potassium: below 3.5 mmol/L
- Hypokalaemia causes widespread downsloping ST depression with T-wave flattening/inversion, prominent U waves and a prolonged QU interval.







↓K^+ ECG hypokalemia- tachyarrhythmias

- Frequent supraventricular and ventricular ectopics
- Supraventricular tachyarrhythmias: AF, atrial flutter, atrial tachycardia
- Potential to develop life-threatening ventricular arrhythmias, e.g. VT, VF and Torsades de Pointes

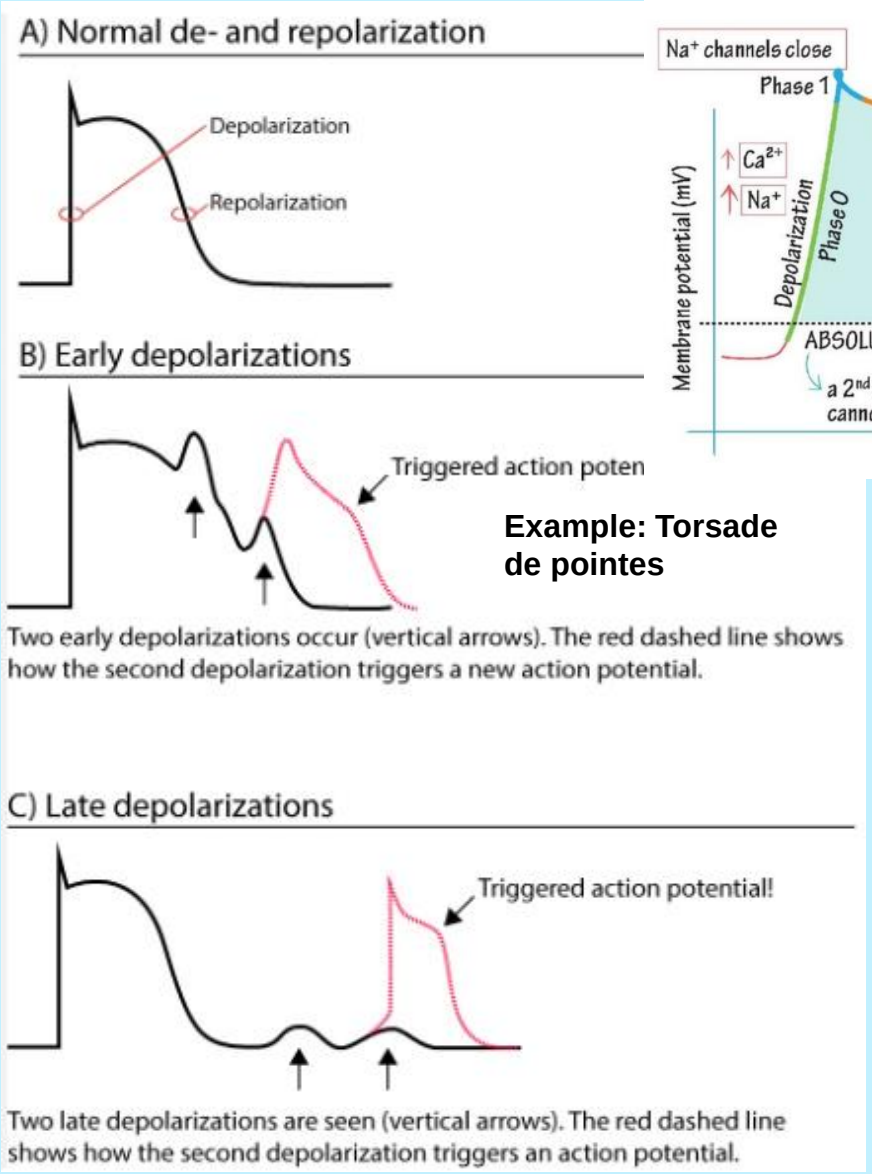


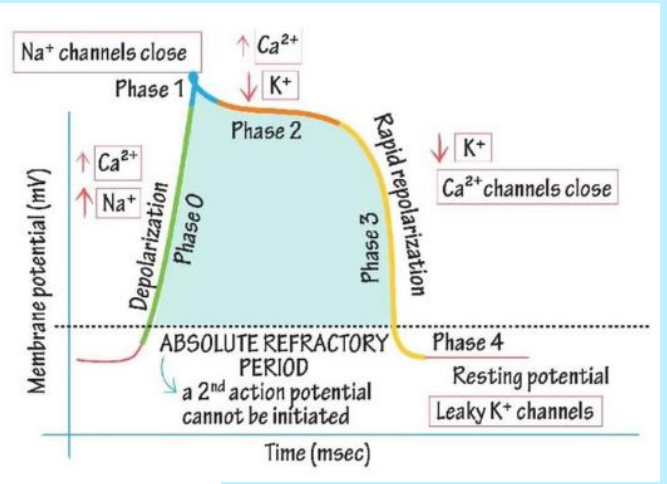
- Hypokalaemia causing Torsades de Pointes: Note the atrial ectopic causing ‘R on T’ (or is it ‘R on U’?) that initiates the paroxysm of TdP

Triggered activity in hypokalemia leading to tachycardia or tachyarrhythmia

- The repeated spontaneous depolarization (without coming impulses) due to instability of membrane potential in the myocytes
- <u>early after depolarization</u>: occur before repolarization has finished (hypokalemia, hypothermia or hypoxia)

- <u>delayed after depolarization</u>: occur after repolarization (from raised intracellular Ca2+)



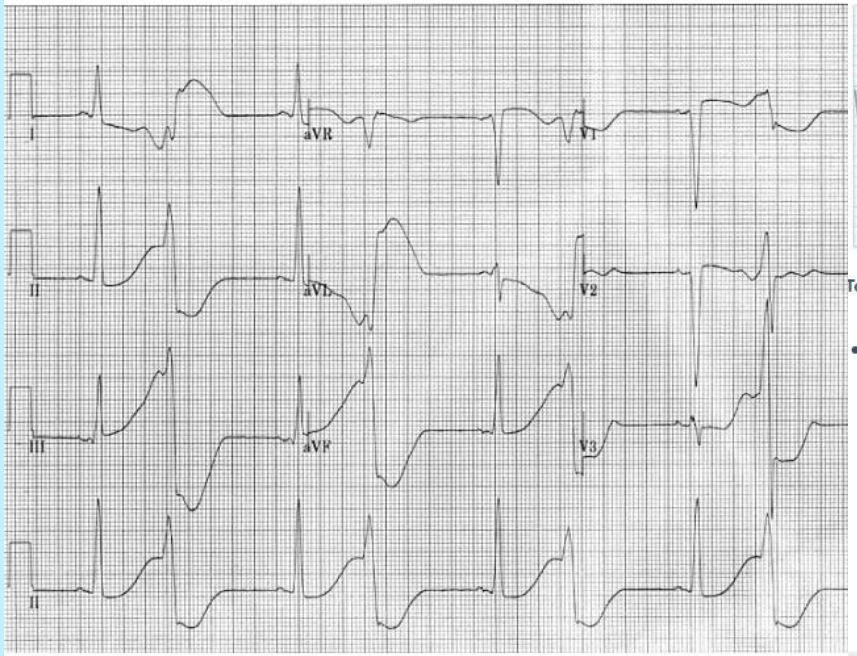


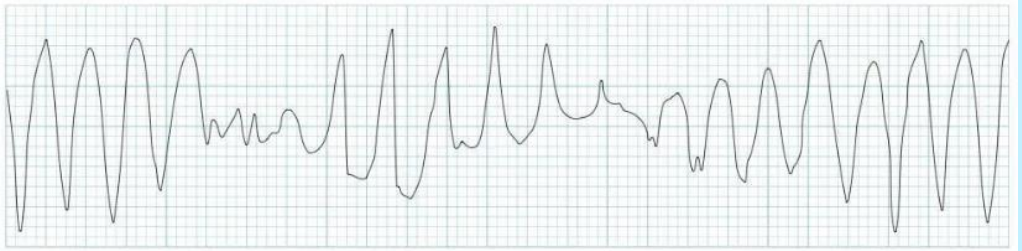
Example: Torsade de pointes

In atrial and ventricular myocytes repolarization due to K^+ efflux: Non-pacemaker action potentials or fast response

Torsade de pointes

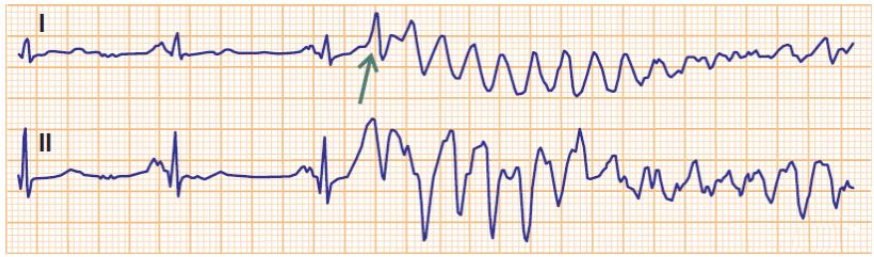
- A specific form of Polymorphic ventricular tachycardia: it has a characteristic morphology in which the QRS complexes “twist” around the isoelectric line. R-on-T phenomenon: ventricular extrasystole stands on descending part of T wave and trigger torsade de points
- Hemodynamic changes, e.g. ↓ cardiac output due to too fast heart beat
- Cause: QT prolongation (inborn defect) or long plateau phase, e.g. early after depolarization

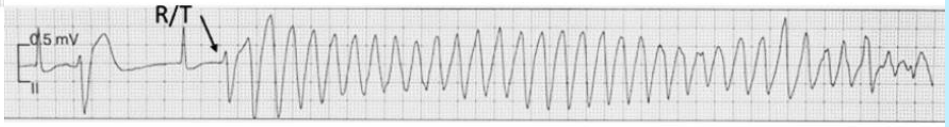




Torsades de pointes

- For TdP to be diagnosed, the patient must have evidence of both PVT *and* QT prolongation



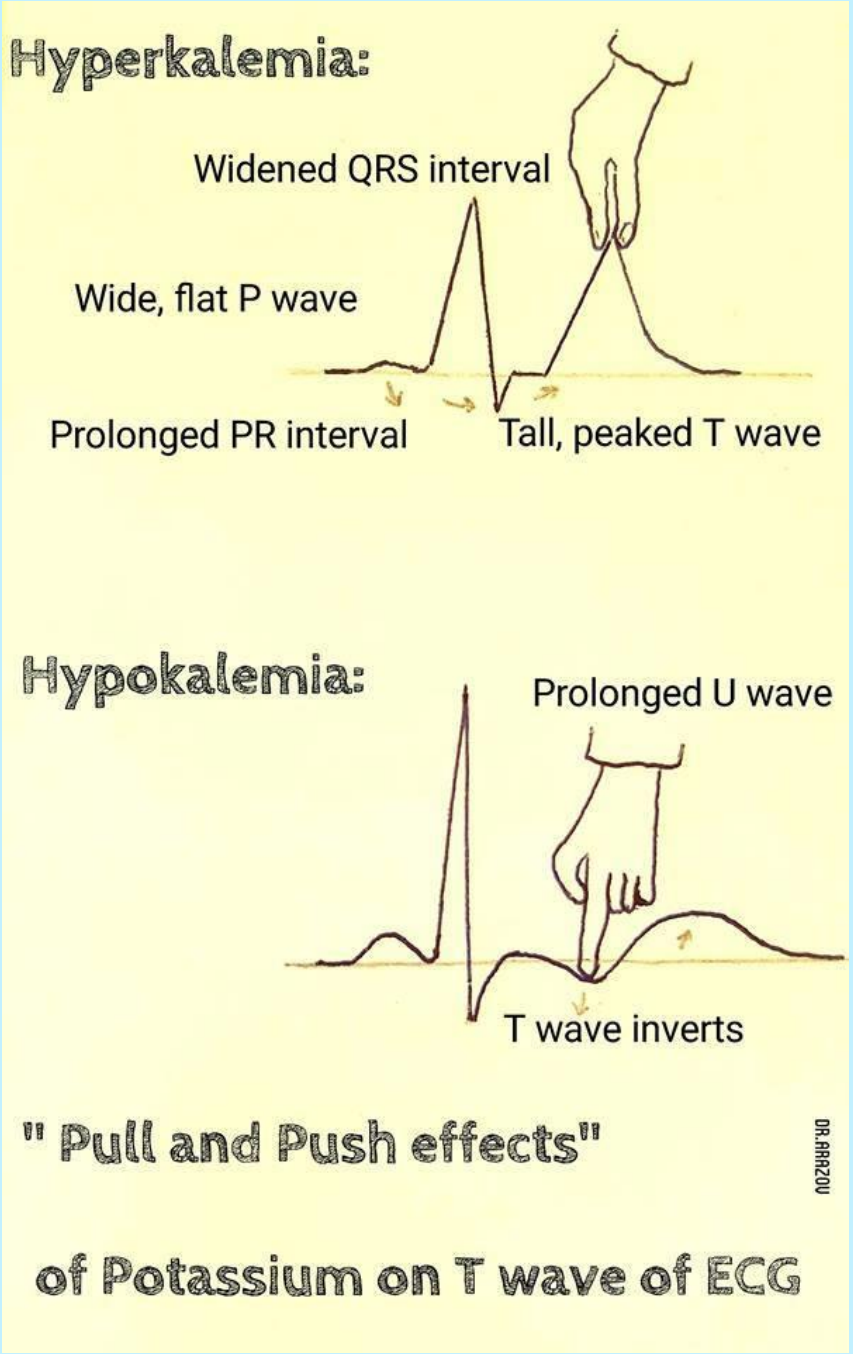


R on T phenomenon:

- There is sinus rhythm with frequent PVCs in a pattern of ventricular bigeminy
- The QT interval is markedly prolonged (at least 600ms), with each PVC falling on the preceding T wave (= ‘R on T’ phenomenon)

↑↓K^+

Comparison of hypokalemia and hyperkalemia on ECG: the first manifestation



Na^+ Hypernatremia and hyponatremia I

Hypernatremia implies a plasma sodium level above
145 mEq/L (145 mmol/L) and characterized cellular dehydration.

- Acute hypernatremia: it has been present for 48 hours or less.
  Fast intervention or the exsiccation of the nerve cells
- Chronic hypernatremia: it has been present for longer than 48 hours.
  Slow intervention or brain edema

Hyponatremia represents a plasma sodium concentration below 135 mEq/L (135 mmol/L) and characterized cellular hyperhydration.

.

- acute hyponatremia: it has developed over a period of less than 48 hours,
  - Fast intervention or fatal herniation (foramen magnum)
- chronic hyponatremia: it has been present for more than 48 hours
  - Slow intervention or osmotic demyelination syndrome

Na^+ Hypernatremia and hyponatremia II

BRAIN ADAPTATION TO severe chronic hypernatremia

- Endogenous/idiogenic osmoles (The brain can produce more particles from its cellular metabolism): taurine, glycine, glutamine, sorbitol and inositol and urea.
- Fast intervention: **cerebral edema**

BRAIN ADAPTATION TO HYPONATREMIA:

- increased intracranial pressure forces interstitial sodium and water out of the brain into the cerebrospinal fluid, ameliorating the increased brain volume. In addition, astrocytes begin to lose intracellular solutes, principally potassium and organic osmolytes.
- Fast intervention: **osmotic demyelination syndrome (ODS)**

Na^+ Hypernatremia and hyponatremia: ECG

- Increased (hypernatremia) and decreased (hyponatremia) sodium levels do not have any effect on the ECG, nor cardiac rhythm or impulse conduction (Non-significant changes on the ECG)

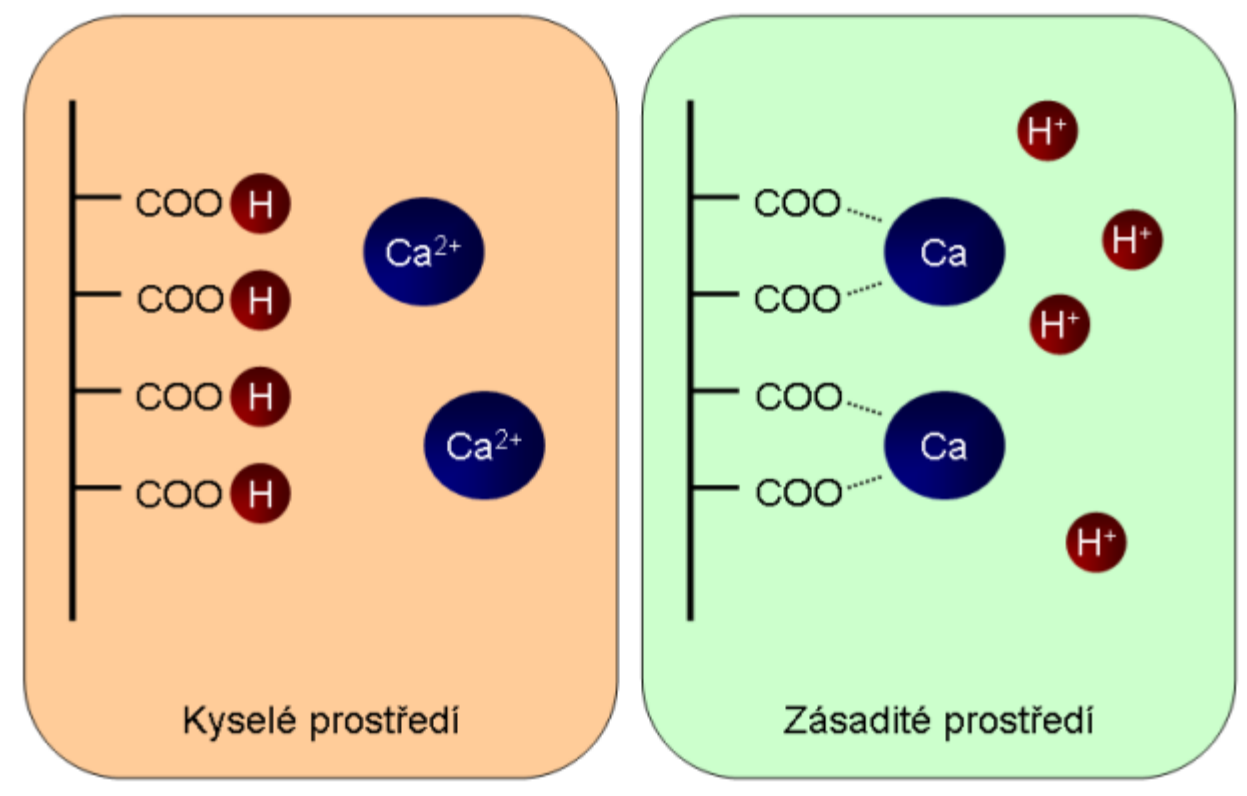
Ca^{+} Calcium

- Most Ca^{2+} (99%): in bone as hydroxyapatite (Ca5(PO4)3(OH))

- Small fraction of Ca^{2+} circulating within the intracellular (primarily in the mitochondria) and extracellular (in the blood):
  - 50% is bound to plasma proteins, (primarily albumin) and
  - 40% is in the free or ionized form (5.5 to 5.6 mg/dl).

Ca^+ The ionized calcium

- In states of acidosis, levels of ionized Ca^{2+} increase.
- In states of alkalosis, the amount of protein-bound calcium increases and the physiologically active, ionized Ca^{2+} level decreases. The decreased concentration of ionized Ca^{2+} may be great enough to cause symptoms of hypocalcemia, such as tetany.



↓Ca^{+} Hypocalcaemia (bellow 2.26 mmol/L): causes

- Nutritional deficiencies
- Excessive amounts of dietary phosphorus also bind with calcium, so neither mineral is absorbed when such an excess occurs.
- Hypoparathyroidism (after removal of the parathyroid glands: loss of PTH) or Pseudohypoparathyroidism
- Vitamin D deficiency
- Blood transfusions: the citrate solution used in storing whole blood binds with calcium and makes it unavailable to the tissues.
- Renal insufficiency: loss of glomerular filtration with phosphate accumulation secondary decreases Ca^{2+} and secondary hyperparathyroidism
- Pancreatitis causes release of lipases into soft tissue spaces, so the free fatty acids that are formed bind calcium, causing a decrease in the concentration of ionized calcium.
- Metabolic or respiratory alkalosis: ionized calcium.
- Hypoalbuminemia: total serum calcium or bound calcium in the plasma.

Hypoparathyroidism

- In the absence of PTH, resorption of calcium from bone and regulation of calcium reabsorption from the renal tubules are impaired.
  - ↓ serum calcium levels
  - ↑ serum phosphate levels

↓Ca^+ Hypoparathyroidism

- Primary:
  - Damage to the parathyroid glands during thyroid surgery and occurs because of the anatomic proximity of the parathyroid glands to the thyroid or ionizing radiation
  - Autoimmune hypoparathyroidism
  - Idiopathic hypoparathyroidism
  - Hemochromatosis: the accumulated iron damages the gland
  - Wilson's disease: the accumulated copper
  - Familial hypoparathyroidism
  - DiGeorge syndrome (velocardiofacial syndrome): thyroid gland malformations
- Secondary:
  - Hypomagnesemia (due to chronic alcoholism, malnutrition, malabsorption) also can cause a decrease in PTH

↓Ca^{+} Pseudohypoparathyroidism

- an inherited condition: There is a resistance to PTH and its receptor cannot produce cAMP in response to PTH

- ↑ PTH
- ↓ hypocalcemia
- ↑ phosphate (hyperphosphatemia).

↓Ca^{+} Hypocalcaemia (bellow 2.26 mmol/L): severe symptoms

- Long-term: Osteoporosis and fractures
- Prolonged ST leads to long QT interval indicating prolonged ventricular depolarization (plateau) and decreased cardiac contractility (inotropic effect), cardiac arrest
- *Tetany:* continuous severe muscle spasm that can interfere with breathing and cause death.
- Intestinal cramping and hyperactive bowel sounds: affected the smooth muscles of the gastrointestinal tract
- an increase in neuromuscular excitability: partial depolarization of nerves and muscle as the threshold potential becomes more negative and approaches the resting membrane potential (hypopolarization): a low threshold for nerve and muscle excitation so that a nerve impulse may be initiated by a slight stimulus
    - Relationship of irritability with cations and anions
    - $Na^{+} + K^{+}/Ca^{2+} + Mg^{2+} + H^{+}$



↑Ca^{+} Hypercalcemia (exceeding 2.65 mmol/l or severe 3.5mmol/L): causes

- Primary hyperparathyroidism: ↑Ca^{2+}
- Secondary hyperparathyroidism : slightly ↑Ca^{2+}
- Tertiary hyperparathyroidism : ↑Ca^{2+}
- Paraneoplastic syndrome: ectopic production of the hormones mimicking the effect of parathyroid hormone: many tumors produce hormones similar to PTH (hypoparathyroidism at parathyroid gland): bone metastases
- sarcoidosis: increase vitamin D levels.
- acidosis: increasing ionized calcium levels.

↑Ca^{+} Primary Hyperparathyroidism

- The calcium level in the blood increases because of increased bone resorption and gastrointestinal absorption of calcium. Normal feedback mechanisms, such as elevated serum levels of ionized calcium, fail to normally inhibit PTH secretion by the parathyroid gland.
  - Parathyroid adenomas
  - Parathyroid hyperplasia
  - Parathyroid carcinoma
  - Multiple endocrine neoplasia syndromes
    - MEN-1: Parathyroid adenomas + gastrinoma (neuroendocrine tumor secreting gastrin, pancreas or duodenum) + prolactinoma (benign tumor of pituitary gland),
  - MEN-2a: carcinoma of thyroid gland + feochromocytom (adrenal medulla) and less hyperparathyroidism
  - Familial hypocalciuric hypercalcemia (FHH)

↑Ca^{+} Familial hypocalciuric hypercalcemia (FHH)

- Benign autosomal dominant condition: It is caused by a mutation in the calcium-sensing receptor in the parathyroid gland.
  - ↑ Ca^{2+}
  - ↓ phosphate (hypophosphatemia)
  - ↓ urine calcium excretion
- It can be differentiated from primary hyperparathyroidism by measurement of 24-hour urine calcium excretion

↑Ca^{+}

Secondary Hyperparathyroidism

- Normocalcemic hyperparathyroidism:
  - Tissue resistance to PTH
  - An early phase of primary hyperparathyroidism

- Secondary to a decrease in serum ionized calcium levels:
  - Chronic kidney disease: failure of glomerular filtration results in hyperphosphatemia and reduced levels of activated vitamin D. Both leads to hypocalcemia, which stimulates PTH secretion.
  - chronic vitamin D deficiency
  - decreased intestinal absorption of vitamin D or calcium

↑Ca^+

Tertiary Hyperparathyroidism

The etiology is unknown but represents autonomous secretion of PTH from persistent parathyroid stimulation.

It occurs:

- after long-standing secondary hyperparathyroidism causing hyperplasia
- after renal transplantation

↑Ca^{+} Hypercalcemia: severe symptoms

- increased cardiac contractility: Short QT interval (short paleau) and ventricular extrasystoles
- decreased in neuromuscular excitability: threshold potential becomes more positive (hyperpolarized) (e.g., moves from −60 to −50 millivolts) and the cell membrane becomes refractory to depolarization because there is a greater difference between threshold potential and resting membrane potential
- Renal tubular function is decreased (polyuria and dehydration), kidney stones form as precipitates of calcium salts
- Osteoporosis, fractures



P

Phosphate

- Phosphate (HPO^{2-}_4): primarily in bone (85%)
- Smaller amounts: within the intracellular (main intracellular anions) and extracellular spaces. In the plasma, phosphate exists
  - phospholipids and phosphate esters
  - inorganic phosphate (ionized form). The normal serum levels of inorganic phosphate range from 2.5 to 4.5 mg/dl and may be as high as 6.0 to 7.0 mg/dl in infants and young children.

Calcium and phosphate

- Calcium and phosphate concentrations are rigidly controlled.
  - [$Ca^{2+} \times HPO^{2-}_4 = K$] constant ($K$)
  - Thus, if the concentration of one ion increases or decreases, that of the other increases or decreases.
- Calcium and phosphate balance is regulated by three hormones:
    - parathyroid hormone (PTH):
      - PTH gene transcription is increased by hypocalcemia, glucocorticoids, and estrogen. PTH release is increased by hypocalcemia, adrenergic agonists, dopamine, and prostaglandin E_2
      - PTH stimulates reabsorption of calcium along the distal tubule of the nephron and inhibits phosphate reabsorption by the proximal tubule of the nephron.
      - The combination of low calcium level and increased PTH secretion causes the renal activation of vitamin D.
    - vitamin D
      - When renal failure occurs, vitamin D is not activated; serum calcium levels decrease; and phosphate levels increase. Chronic kidney disease (decreased glomerular filtration leads to P accumulation that secondary decrease Ca^{2+})
    - calcitonin.

hormone	response to	serum	bone	GIT	kidney
Parathyroid hormone parathyroid gland	↓ Ca^{2+}	hypercalcemia and hyperphosphatemia	↓	via calcitriol	↑ reabsorption of Ca^{2+} but excretion of P and HCO_3^-
Calcitriol (from cholesterol in the skin, than kidney and liver)	parathyroid hormone	Depends on concentration		↑absorption of Ca^{2+} and P in same ratio	↑ reabsorption of Ca^{2+} and P
Calcitonin thyroid gland	Acute (not chronic) hypercalcemia	hypocalcemia	short-term bone calciphication	↓absorption of Ca^{2+} and P in same ratio	↑excretion of both Ca^{2+} and P

↑**P** Hyperphosphatemia (more than 1.46 mmol/L): causes

- loss of glomerular filtration: ↑P secondary decreases Ca^{2+} that and secondary hyperparathyroidism
- the cell destruction associated with treatment of metastatic tumors with chemotherapy: most phosphate is located in cells
- Increased somatotropin (growth hormone): acromegaly (diseases of the Anterior Pituitary) and kids
- Hypoparathyroidism can cause elevated phosphate levels by increasing renal tubular reabsorption of phosphate.
- High D vitamin

↑**P**

Hyperphosphatemia: Symptoms

- are related primarily to low serum calcium levels and thus are comparable to symptoms of hypocalcemia.
- With prolonged hyperphosphatemia, calcification of soft tissues occurs in the lungs, kidneys, and joints because ↑ P with Ca^{2+} are deposited in bone and soft tissues.

↓P Hypophosphatemia (less than 0.81 mmol/L): causes

- decreased tubular reabsorption within the proximal tubule: leads to
  - Kids: rachitis (rickets) resistant to vitamin D
  - Adults: idiopathic osteomalacia (Milkman syndrome)
  - De Toni-Fanconi syndrome: decreased phosphate reabsorption plus glucose and amino acids
- long-term alcohol abuse
- malabsorption syndromes: decreased reabsorption of phosphate
- Hyperparathyroidism
- D vitamin deficiency

↓**P**

Hypophosphatemia: Symptoms

- ATP levels
- reduced capacity for oxygen transport:decreased 2,3-diphosphoglycerate and the oxyhemoglobin curve shifts to the left, hypoxia can occur with bradycardia
- bone resorption occurs

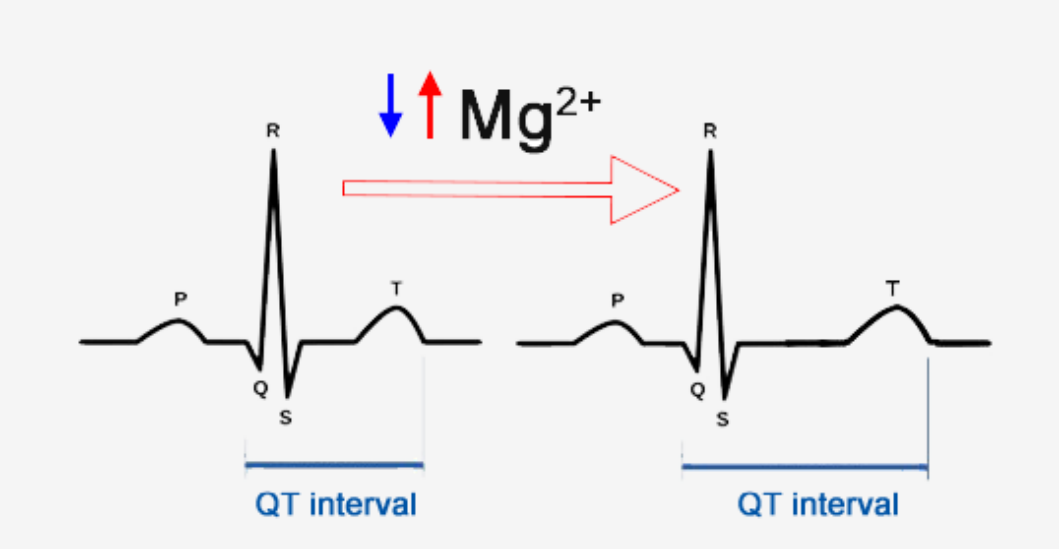
↓Mg^{2+} Hypomagnesemia <0.7mmol/L

- Cause: severe pancreatitis (The intestinal fat /steatorrhea can bind Mg^{2+} being similar to Ca^{2+}), alcoholism, diuretics or diarrhea

- Manifestation:
  - A loss of magnesium results in decreased parathyroid hormone and hypocalcemia
    - Similar to hypocalcemia, a low Mg (<0.5mmol/L) increases neuromuscular excitability and tetany are present. ECG: long QT interval that increases the risk of Torsades de Pointes
  - A loss of magnesium results in movement of potassium out of the cell, with renal excretion resulting in hypokalemia.

↑ Mg^{2+} Hypermagnesemia > 1 mmol/L

- It is rare except for renal failure or cellular destruction
- ECG: long QT interval (same as low Mg2+) + long PQ



References

Look up the particular theme in the following indexes

- Porth's Pathophysiology: Concepts of Altered Health States (9th Edition) by Sheila Grossman, Carol Mattson Porth. Wolters Kluwer Health | Lippincott Williams & Wilkins, 2014.
- McCance K. L., Huether S. E.: Pathophysiology. 8th Edition. Mosby, 2018.
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